AI Campus Route Navigator

Assignment X_03 — Refactored Brief Report

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1. Objective

The project models the CU Technology Campus as a weighted graph and compares PATHFINDER (Greedy Best-First Search) with ORBIT (A* Search). The comparison records route, path cost, explored states, and execution time.

2. Graph and file handling

Campus locations, approximate coordinates, connections, and walking costs are stored in campus.json. The graph loader validates node references and positive edge costs, keeping campus data separate from routing logic.

3. Search state and CSE constraint

The search state is (location, mode, previous_location). The CSE zone contains CSE Laboratory, CSE_AKC Seminar Hall, and CSE_Reflexon Room. Entry requires Tower 2 Front/Rear Entry → Lift Area → CSE location. Inside CSE, only CSE locations and the Lift Area are allowed. Exit requires CSE location → Lift Area → Tower 2 Front/Rear Entry → ordinary campus. A start inside CSE is initialized in CSE mode.

4. Heuristic

Both algorithms use h(n) = 0.55 × sqrt((x_n-x_d)^2 + (y_n-y_d)^2). The coordinates are approximate map coordinates and the scale factor is conservative relative to the stored edge costs.

5. Algorithms

PATHFINDER prioritizes h(n), so it focuses on estimated closeness to the destination. ORBIT prioritizes g(n)+h(n), combining accumulated route cost with estimated remaining cost.

6. Refactoring

Responsibilities are separated across config.py, models.py, campus_map.py, search.py, agents.py, experiment.py, main.py, and test_search.py. The refactor adds type hints, validation, immutable data models, pathlib-based file handling, reusable search logic, and regression tests.

7. Conclusion

The experiment demonstrates the difference between heuristic-only search and A*. It also shows why the CSE routing requirement needs contextual search state rather than treating every physical location as an unconstrained graph node.

8. Modelling note

Graph connectivity, coordinates, and walking costs are modelling estimates derived from the supplied map. They should be verified against the assignment map before submission.

9. Current experiment results

Source → Destination	Agent	Cost (m)	Nodes	Time (s)
Reception of Calcutta University → Library	PATHFINDER	475.0	5	9.3e-05

Source → Destination	Agent	Cost (m)	Nodes	Time (s)
Reception of Calcutta University → Library	ORBIT	475.0	10	6.2e-05
Canteen, CU Technology Campus → New Building 2 (Workshop Building)	PATHFINDER	290.0	3	1.8e-05
Canteen, CU Technology Campus → New Building 2 (Workshop Building)	ORBIT	290.0	3	1.5e-05
Entry Gate 1 (G1) → CSE Laboratory	PATHFINDER	685.0	14	7e-05
Entry Gate 1 (G1) → CSE Laboratory	ORBIT	685.0	40	0.000166
Auditorium Hall → CSE_AKC Seminar Hall	PATHFINDER	485.0	9	4.8e-05
Auditorium Hall → CSE_AKC Seminar Hall	ORBIT	485.0	33	0.000135
Library → Canteen, CU Technology Campus	PATHFINDER	440.0	19	7.8e-05
Library → Canteen, CU Technology Campus	ORBIT	440.0	10	5e-05
CSE_Reflexon Room → Entry Gate 4 (G4)	PATHFINDER	570.0	7	2.8e-05
CSE_Reflexon Room → Entry Gate 4 (G4)	ORBIT	570.0	9	3.1e-05

Timing values vary by machine and Python runtime; they are included as indicative measurements.